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Energy Prediction and Optimization based on Machine Learning

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Abstract. The paper proposes energy prediction techniques on Electric energy consumption from existing electricity supply using machine learning. Machine learning and its techniques are vastly used in factories for maintenance, in industries for optimization and to calculate the fault tolerance to produce an efficient monitoring system. This paper proposes the best model to monitor the usage of power for predicting the energy Raspberry Pi 3Model B+ is the core controller used for this energy prediction and modelling. AC-current, AC-voltage and power are the input parameters measured using current sensor (ACS712) and AC-Voltage sensor (ZMPT101B). MCP3008 ADC is interfaced with the controller to read analogue current and voltage values. Linear regression and logistic regression are used for both prediction and optimization. The two algorithms are implemented on the Spyder IDE which is installed in the Raspberry pi 3Model B+ controller. After reading the input parameter it predicts the energy consumption and displays whether it is high or low for that day. Experimental Results suggests that the linear regression model is the best among the two because the accuracy metric values viz., Mean Absolute Error, R2 score, Execution time, Residual Sum of Square are comparatively less when compared with logistic regression model.

Keywords: Regression, Logistic, Machine Learning, ACS712, ZMPT101B, MCP3008, Raspberry Pi 3Model B+, Spyder IDE

INTRODUCTION

Mapping the electrical energy consumption level with the electricity supply level is not an easy process, because the extra electricity produced cannot be stored, until it gets converted into other forms which require additional budget and resources. Concurrently, miscalculating consumption of energy could be tragic, with extra demand even blackout could be possible when the supply line is overloaded. With the arrival of machine learning the energy required in future could be calculated precisely by having current electric energy consumption data in hand. There are two main advantages when the prediction is accurate. First, one can get the key factors that affect the energy demand in their building and providing an opportunity to improve the energy efficiency. Second, to alert the manager of the building when the electricity consumption is low/high. This paper implies on developing a prediction unit using two algorithms: 1) Linear regression [6] and 2) Logistic Regression [7] and then suggest a best model for optimization. When the input parameters are given, then the designed prediction unit predicts the energy which is the output by making use of the dataset [8] which has the measured values. Then the optimization part takes place with the suggested model. Both the prediction and optimization part take place in the Spyder IDE in the Raspberry Pi controller. Then time and energy values are taken out as a zip file which has a spreadsheet that can be opened with Libreoffice software in Raspberry pi for creating charts that shows the peak energy consumption time.

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RELATED WORKS

Xinyue Cuiet et al (2019) proposed a prediction algorithm based on adaptive filter and a node clustering technique to overcome the issues faced in WSN that affects its application on smart cities [1].The proposed mechanism shows good results on energy harvesting in wireless sensor network. The authors of this paper use light energy as an additional energy source. The negative thing is this technique could not be applicable in underwater and in some areas where the light could not reach. So, we need to find an alternate renewable source of energy as supplement energy in those areas where light energy could not be used.

Smrutishikta Das et al (2019) designed an AI based energy prediction technique for energy consumption based on data analytics to increase its efficiency [2]. A vast level of historical data analytics in addition with a collection of modelling algorithm to interpret every logical part of that analytical data is required for the AI based smart predictor.Once the data pre-processing gets completed then linear regression is implemented to predict the consumption of energy per carpet area. The performance of the prediction gets raised up to 31.03% when the manual evaluation takes place. The main flaw of this paper is the manual calculation and data visualization.

Ronald van Leeuwen et al (2020) has developed an experimental setup to calculate the ratio error in voltage dependence and phase displacement of current transformers (CTs) that are meant to use in high-voltage (HV) or medium voltage applications [3]. Current transformer traces only the primary current that passes through a shield of High Voltage cable with shield to the GND. The main drawback of this setup is we need to have a greater number of measurements and, they need to be processed to display the current dependence of the results, particularly at higher currents the relative effect of the leakage current supposes to be less.

Shengyuan Zhong et al (2019) proposes an analysis for power consumption and interpreted the differences between the operation and planning stage through mutual information [4]. Designing an energy system which also contains renewable energy is the best way to save energy, reduce electric energy consumption and can reduce emission. Somehow the energy consumption in the building during the operational stage gets deviated from the calculated values of planning stage. The planning model has the control of renewable energy part in the building energy system. The limitation of this paper is the actual operation process would deeply modify the consumption of energy in the building energy system due to the lack of strong interaction between the demand and supply information.

Xiangyu Li and Nijie Xie (2018) proposes an energy prediction structure using multi-model fusion for energy harvesting in Wireless Sensor Network node [5]. In this paper the author suggests an accurate and low overhead prediction method for energy harvesting on WSN nodes which is mandatory for proper management of power. The prediction error was reduced to a greater extent because of implementing solar radiation prediction model and even it reduces the energy required by a battery to recharge it by almost 26.4%. The scope of this paper is low because of the complexity in implementing multiple prediction model and it is applicable only for light-weight Micro Controller Units, e.g., MSP430.

METHODOLOGY

Figure 1 shows the block diagram of the proposed methodology which is designed using aRaspberry Pi 3Model B+ controller, Current sensor (ACS712), Ac-Voltage sensor (ZMPT101B), MCP3008 Analog to Digital Converterand with essential software tools. The prediction unit uses a deep learning technique. They are developed in Raspberry Pi controller using Spyder IDE software that uses python programming language. The energy is predicted from the created dataset which has the input parameters that are measured and listed through the designed setup. The designed setup comprises of two sensors, current and voltage sensor which are interfaced with the MCP3008 analogue to digital converter, because the Pi controller cannot read the analogue values. Then the Raspberry Pi controller is interfaced with the MCP3008 channels which reads the analogue values from the sensors. Then the two algorithms linear regression and logistic regression are implemented on the Spyder IDE that is installed in the Raspberry Pi controller. Finally, the predicted energy and the peak power consumption time are displayed as shown in (Fig. 9 and Fig.10). The prediction unit workflow is diagrammatically represented on Fig. 2.

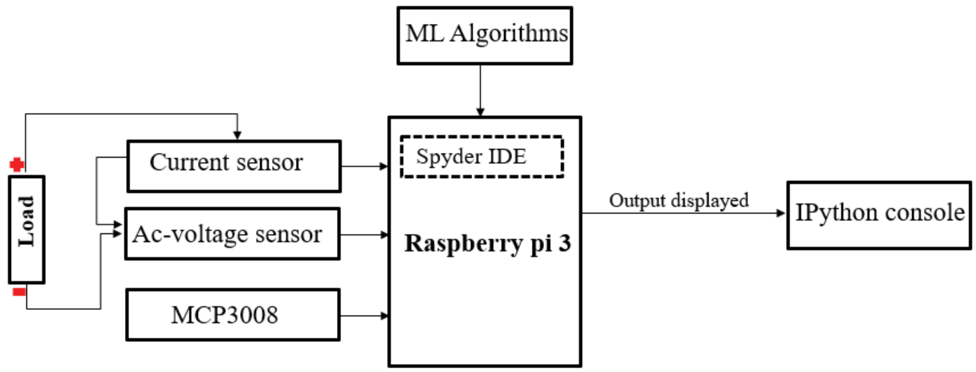

FIGURE 1. Block diagram of the methodology

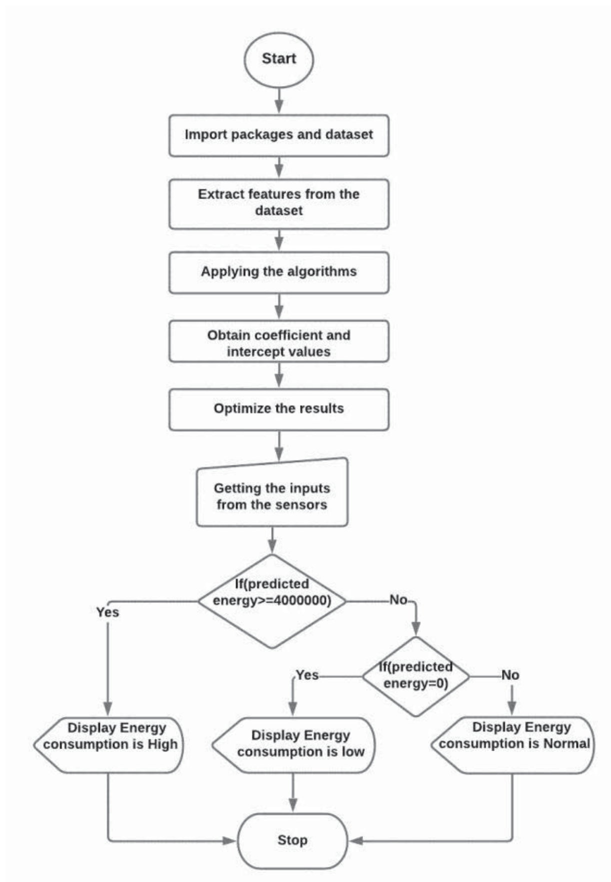

FIGURE 2. Flow chart of the prediction unit

EXPERIMENTAL RESULTS

The prediction unit of the Raspberry Pi controller is designed with python programming language on Spyder IDE where the logistic regression and linear regression is implemented, and the accuracy metric is calculated. For the two algorithms a different epoch is set to improve the loss and accuracy metrics. The obtained results are optimized using the RMS (Root Mean Square).The various stage involved in designing the prediction unit, optimization techniques and calculating the accuracy are explained below.

Stage 1. Implementing two algorithms and optimization technique

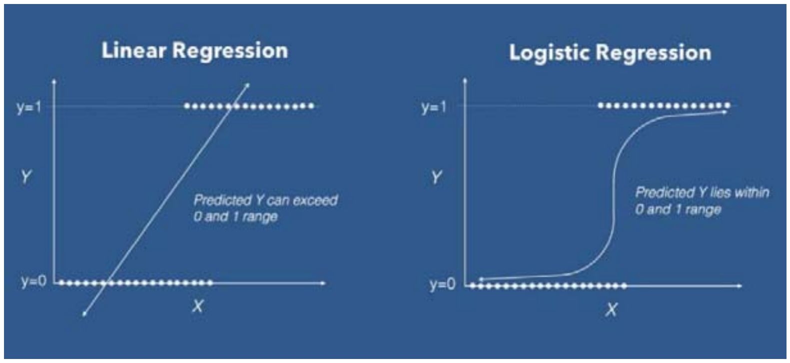

FIGURE 3. Linear and Logistic regression (Source: [10])

The created dataset using the measured values from the sensor is converted into .csv format, so that it can be imported in the python programming using panda's package. df.mask () function is used to split the dataset and for testing and training. About 80% of the data is used for training and 20% of the data is used for testing. The sklearn [9] is the library from which (both the linear and logistic) regression model is imported from and input parameters extracted from the labelled dataset is current, voltage, power and energy is set as the output.

$$Y = B0 + B1*Xi1 + B2*Xi2 + Bn.Xin \qquad (1)$$

where Y, Xi are dependent (Energy value) and independent variable(current, voltage and power) and **B0, Bn** is intercept and slope values. The coefficient and intercept values are determined using the dataset's parameters, with the input label set to Xi andthe output label set to Yi. Pseudo code of the program is mentioned below.

1. Start with Iiporting library files
2. Read Number of Data (n) and initialize inputs as dataframe then train and test the dataframe
3. Define linear / logistic model
4. Assign trained values to x and y andfitting x and y and calculate Required Constant a and b
5. Print intercept (a) and coefficient (b) value
6. Substitute the value of a and b in y= a + bx which is required line of best fit
7. Testing x and y and Print MAE, RSS, execution time and R2-score
8. Assigning epochs value and finally print voltage, current, power, predicted value and end.

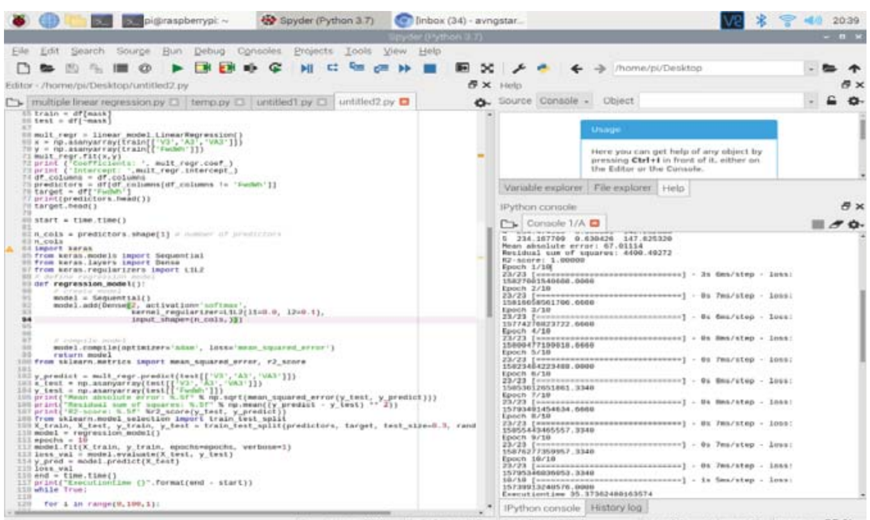

FIGURE 4. Logistic regression is implemented on Spyder IDE

The Fig. 4 shows the accuracy metrics like Mean Absolute Error, R2-score, Residual Sum of Squares and execution time when Logistic regression model is implemented in Spyder IDE.

The accuracy metrics obtained for linear regression and logistic regression is displayed in the Table 1.

TABLE 1: Comparison Results

	Mean absolute error	**Residual sum of squares**	**R2-score**	**Execution time**
Multiple linear regression	37.27219	1389.21634	1.00000	14.97063660621sec
Logistic regression	67.01114	4490.49272	1.00000	35.37362480163sec

Stage 2. Error calculation and evaluation

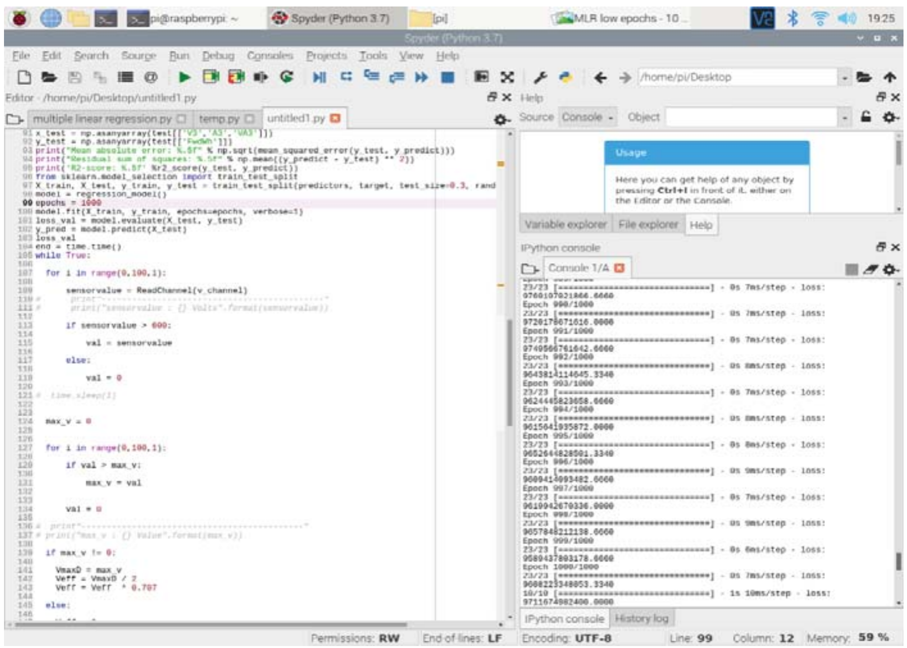

FIGURE 5. Performance model of linear regression at 1000 epochs

The average square loss is the regression based basic loss function that computes the distance between the target values and the expected values. It is implemented by importing the model. Compile () function which present in the keras library. The error is evaluated by repeating the process for different set of epochs values for those two algorithms. Figure 5 shows the performance of linear regression at 1000 epochs.

The error value is evaluated at different set of epochs for the two machine learning algorithms, and it is displayed in the Table 2.

TABLE 2. Comparison of Error Values

	Epochs	**Mean Square error (Loss)**
Multiple linear regression	50	15705634242560.0000
	100	15678673256448.0000
	500	13393796792320.0000
	1000	9711674982400.0000
Logistic regression	50	15739957280768.0000
	100	15739946795008.0000
	500	15739792654336.0000
	1000	15739619639296.0000

The linear regression model has best accuracy metrics than logistic regression model. It has the least Mean Absolute Error, Residual sum of squares and considerably low error. The optimization and overall performance development of the two models at different epochs is shown in the Fig. 6.

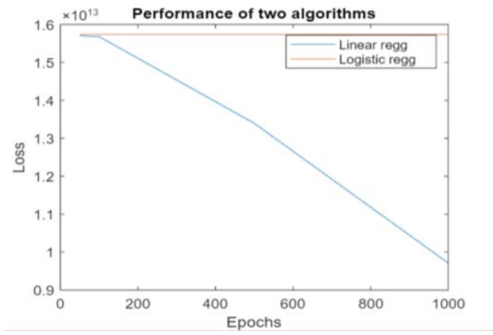

FIGURE 6. Overall performance of two models

Stage 3. Configuration and setup

Figure 7 shows the hardware setup of the proposed system. The input from the current and voltage sensor are given to Raspberry Pi which runs the prediction algorithm and calculates the output by using the predicted coefficient and intercept values form the dataset. The predicted output i.e., the amount of energy going to be consumed for that day and also calculates if the energy consumption is going to be high or low for that day will be displayed in the IPython console of the Spyder IDE. Finally, the peak energy consumption time will be shown in the graph.

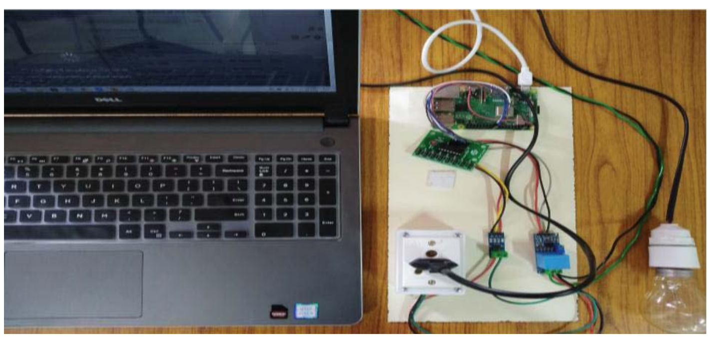

FIGURE 7. Hardware setup

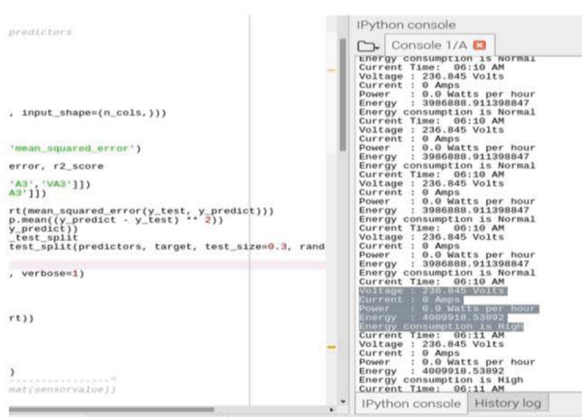

FIGURE 8. Energy consumption from low to normal

FIGURE 9. Energy consumption from normal to high

The Fig. 8 shows the time at which the energy consumption turns from low to normal when implementing the linear regression at 1000 epochs. Similarly, Fig. 9 shows the point at which the energy consumption turns from normal to high.

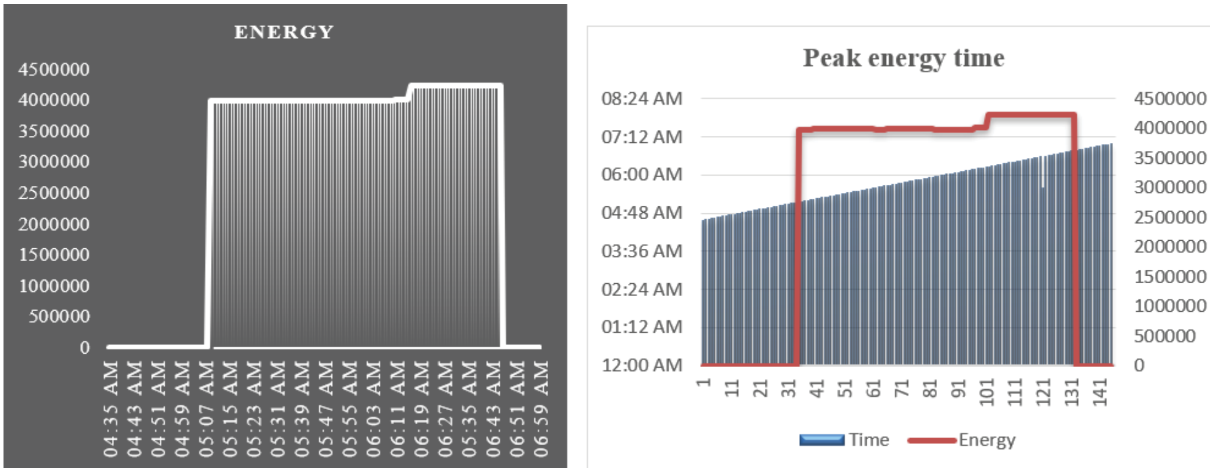

FIGURE 10. Peak power consumption time

The Fig.10 shows the peak energy consumption time that is drawn with the reference of results obtained from implementing linear regression model with epochs set at 1000 in Spyder IDE which is show in Fig.9 and 10. From the Fig.10, it is clear that the peak energy consumption time starts at 06:11 AM and ends at 06:44 AM.

CONCLUSION

The existing system used a prediction mechanism to solve the power issues faced by WSN. In this paper the proposed solution can be used in any power consumption areas or industries to predict usage of energy and give results of peak power consumption time. The energy prediction method was proposed, and an optimization technique was suggested to evaluate the output values. By using linear regression and setting the epochs at 1000 an optimized result was obtained, and the accuracy metrics were vastly improved. The linear regression model has the best accuracy when compared with logistic regression with less mean square error and with the best mean absolute error of 37.27219, a residual square value of 1389.2163 and execute the process with lesser execution time of 14.970sec. Though the R2 score values are same the other metrics of the linear model is two times more accurate than the logistic model. The peak energy consumption time is also described graphically in Fig.10.

This paper can be extended further by adding a renewable energy source like solar energy that can be used as a supplementary energy source. When there is a chance of high energy consumption then the main connection of a building can draw that extra energy from the supplementary source. By doing we can save the money and the energy demand is also controlled.

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